

Lem serie fourier derivada

Lema 1. Sea $f: \mathbb{T} \rightarrow \mathbb{C}$ en $\mathcal{C}^m(\mathbb{T})$ para algún $m \in \mathbb{N}$, entonces

$$\forall n \in \mathbb{Z} \setminus \{0\} : \widehat{f^{(m)}}(n) = (2\pi in)^m \widehat{f}(n).$$

Por tanto, $\lim_{|n| \rightarrow \infty} |n|^m |\widehat{f}(n)| = 0.$

Demostración: Vamos a demostrar el caso $m = 1$, el resto se demuestra por inducción.

Sea $f \in \mathcal{C}^1(\mathbb{T})$, entonces por integración por partes tenemos que $\forall n \in \mathbb{Z} \setminus \{0\}$:

$$\widehat{f'}(n) = \int_0^1 f'(x) e^{-2\pi inx} dx \stackrel{\text{IPP}}{=} f(x) e^{-2\pi inx} \Big|_0^1 + 2\pi in \int_0^1 f(x) e^{-2\pi inx} dx = 2\pi in \widehat{f}(n).$$

Para la segunda parte, de la relación $\widehat{f^{(m)}}(n) = (2\pi in)^m \widehat{f}(n)$ obtenemos

$$|\widehat{f}(n)| = \frac{|\widehat{f^{(m)}}(n)|}{|2\pi in|^m} = \frac{|\widehat{f^{(m)}}(n)|}{(2\pi)^m |n|^m} \implies |n|^m |\widehat{f}(n)| = \frac{|\widehat{f^{(m)}}(n)|}{(2\pi)^m}.$$

Como $f^{(m)} \in \mathcal{L}(\mathbb{T})$ por ser continua en un compacto, por el Lema de Riemann-Lebesgue sabemos que $\widehat{f^{(m)}}(n) \xrightarrow{|n| \rightarrow \infty} 0$, de donde se sigue que

$$\lim_{|n| \rightarrow \infty} |n|^m |\widehat{f}(n)| = \lim_{|n| \rightarrow \infty} \frac{|\widehat{f^{(m)}}(n)|}{(2\pi)^m} = 0. \quad \blacksquare$$

Referenciado en

- prop-clase-ck-velocidad-convergencia-uniforme-fourier

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